# Supplementary Material for "Leakage, Shift and the Operating Point: Evaluating Blood-Smear Leukemia Classifiers Across Time and Laboratories"

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This document contains the complete settings, computing environment, dataset details, full parameter sweeps, secondary analyses and the TRIPOD+AI checklist referred to in the main text. Code, split files, the pre-specification and per-model outputs are released with the paper.

## S1. Settings and computing environment

TABLE S1: Complete settings of every component. All values were fixed in advance or taken from library defaults; none was fitted or tuned on an external set, except that the preprocessing selection rule used external performance on a subset that included Validation patients (disclosed in the main text). Defaults of the named libraries apply to anything not listed.

| Component | Setting | Value |
|---|---|---|
| Data and splits | Development / external cohorts | APL-AML: Discovery (82 patients) / Validation (24 patients), split by diagnosis date by the data owners |
| | Second dataset | C-NMC 2019: official folds 0–1 (42 subjects) / fold 2 (31 subjects), designated before any analysis |
| | Cross-validation | 5 folds; image-level `StratifiedKFold` or patient-level `StratifiedGroupKFold` (cells), `StratifiedKFold` over patients (T3) |
| | Random seeds | 0, 1, 2 (fold assignment, cell sampling, model initialization); results are averaged over seeds |
| | Cells per patient (T3) | Up to 500, sampled at random once (seed 0); patients with fewer cells keep all |
| Engineered features (93) | Input | RGB, bilinear resize to $192 \times 192$ |
| | Color deconvolution | Ruifrok–Johnston H&E matrix (`skimage rgb2hed`) |
| | Nucleus | Gaussian smoothing ($\sigma = 1$) of hematoxylin, Otsu threshold, opening (disk 2), component nearest the center (within $0.30\times$ image size); fallback: darkest 15% of gray levels |
| | Cytoplasm proxies | Basophilic region: blue-dominant pixels within one nuclear radius, opening (disk 2), closing (disk 4); ring: 2–9 px from the nucleus, excluding pixels brighter than the 90th lightness percentile $-3$ |
| | Texture | GLCM on 33 gray levels, distances 1, 2, 4 px, angles $0, \pi/4, \pi/2, 3\pi/4$ (contrast, dissimilarity, homogeneity, energy, correlation; averaged over angles); uniform LBP with $(P, R) = (8, 1)$ and $(16, 2)$ |
| | Missing values | Unreadable images give NaN rows, imputed with the training-fold median |
| | C-NMC background | Pixels with brightest channel $< 15$ set to white before feature extraction (engineered features only) |
| Deep features | Encoder | EfficientNet-B0, ImageNet weights (`timm`), frozen, global-average-pooled 1,280-d output |
| | Input | Bicubic resize to $224 \times 224$, ImageNet mean/std normalization, no augmentation |
| | Extraction | Batch size 64, 2 data-loader workers, CPU, `torch.no_grad` |
| Feature engineering | PCA | Standardization then PCA (`svd_solver`=auto, `random_state`=0), both fitted on training-fold cells only; 16, 32, 64 components pre-specified; 4, 8, 128, 256 as sensitivity analysis; pooled component statistics divided by one common scale (no whitening); per-component standardization reported for comparison |
| | Correlation filter | Drop the later feature of every pair with $\lvert r\rvert > 0.9$ (training cells, $\leq$20,000 sampled) |
| | Stain normalization / L1 | Per-slide median/SD standardization of the 26 stain features; L1-penalized LR, $C = 0.5$, SAGA solver |
| Cell-level classifiers (T1, T2) | Logistic regression | L2, $C = 0.1$, L-BFGS, 3,000 iterations, class-balanced weights; median imputation and standardization inside the pipeline |

| Component | Setting | Value |
|---|---|---|
| | LightGBM | 400 trees, learning rate 0.05, 31 leaves, row subsampling 0.8 (every iteration), column subsampling 0.8, class-balanced weights, 2 threads |
| | Calibration | Temperature scaling fitted on an inner 80/20 split of the training fold (bounded search 0.05–20) |
| Pooled patient model (T3) | Patient vector | Mean, SD, 10th, 50th, 90th percentile of every cell feature ($5\times$ dimensions) |
| | Classifier | LR, L2, $C = 0.1$, 5,000 iterations, class-balanced weights, standardization inside the pipeline |
| Gated attention MIL (T3) | Architecture | Linear($d \to 128$)–ReLU–Dropout(0.25); gated attention with $\tanh$ and sigmoid branches of 64 units; linear classifier on the attention-weighted mean |
| | Optimization | AdamW, learning rate $2 \times 10^{-4}$ (constant, no schedule), weight decay $10^{-3}$, one bag per step, 40 epochs, no early stopping |
| | Loss | Binary cross-entropy with positive-class weight $n_{\mathrm{AML}}/n_{\mathrm{APL}}$ of the training fold |
| | Augmentation | Random removal of 20% of a bag's cells at every step (bags with more than 10 cells) |
| | Input scaling | Median imputation and standardization from training-fold cells |
| Fine-tuned CNN (T1, T2) | Trainable part | EfficientNet-B0 stages 6–7, $1 \times 1$ head convolution (1,280 channels), new linear classifier (dropout 0.2); stem and stages 1–5 frozen, their $112 \times 14 \times 14$ outputs cached once (float16) |
| | Optimization | AdamW, peak learning rate $3 \times 10^{-4}$, 1 warm-up epoch then cosine decay, weight decay $10^{-4}$, batch 128, 6 epochs, class-weighted binary cross-entropy |
| | Data | Up to 5,000 training cells per fold (random); 15% of training patients (grouped split) or images (image split) as inner validation for early stopping by validation loss; one seed |
| | Augmentation | Random horizontal and vertical flips and $90^\circ$ rotations of the cached feature maps |
| Re-run baselines | B1, B2 | Promyelocyte fraction (ranked directly); LR on the 17-class differential count |
| | B3 | EfficientNet-B0 + RBF-SVM ($C = 1$, $\gamma$ = `scale`, Platt probabilities, class-balanced, 8,000 training cells), cell scores averaged per patient |
| Operating point and statistics | Threshold | Highest threshold giving sensitivity $\geq 0.95$ on out-of-fold Discovery predictions; Youden threshold reported for reference |
| | External prediction (T3) | Mean of the 15 fold models (5 folds $\times$ 3 seeds) |
| | Confidence intervals | Patient-level bootstrap, 2,000 resamples (1,000 for cell-level CV); paired differences 4,000 resamples |
| | Calibration error | ECE with 15 equal-width bins |
| | Multiplicity | Holm correction across the comparisons reported in a table |
| Additional analyses | Pooled CV (APL-AML) | 3 repeats $\times$ 5-fold patient CV over all 106 patients, stratified by diagnosis and cohort; threshold from an inner 3-fold CV |
| | Local recalibration | $k \in \{2, 4, 6, 8\}$ local Validation patients ($k/2$ per class), 2,000 draws; threshold lowered to just below the lowest local APL score |
| | Decision curves | Platt recalibration on Discovery out-of-fold scores, prior shift to 10% prevalence, threshold probabilities 0.01–0.60 (the figure in the main text shows 0–0.30) |

TABLE S2
COMPUTING ENVIRONMENT AND RUN TIMES.

| Item | Specification |
|---|---|
| Hardware | Intel Xeon 2.80 GHz, 2 vCPUs, 7.8 GB RAM, no GPU |
| Software | Ubuntu 24.04.5 LTS; Python 3.13.16; PyTorch 2.14.1 (CPU); scikit-learn 1.9.1 [1]; LightGBM 4.7.0; scikit-image 0.26.0; timm 1.0.30 |
| Random seeds | 0, 1, 2 |
| Engineered features | 16.1 min for 25,724 cells (2 processes) |
| EfficientNet-B0 embeddings | 51.2 min for 25,724 cells (2 threads) |
| Cell-level model, 5-fold CV | 11–40 s (LR), 15–600 s (LightGBM) |
| Attention MIL, 5-fold CV | 33–61 s per seed |

## S2. Datasets, model overview, evaluation protocols and segmentation

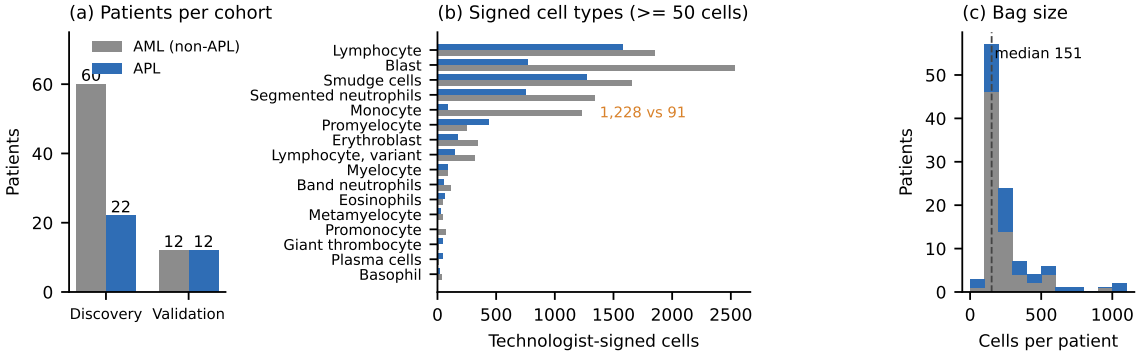



Fig. S1. APL-AML cohort composition. (a) Patients per cohort and diagnosis. (b) Labeled cell types with at least 50 cells; monocytes are almost absent in APL (1,228 vs. 91 cells). (c) Number of cells per patient.

TABLE S3
Composition of the C-NMC 2019 data used here (official training folds).

| Set | Class | Subjects | Cells | Cells per subject, median (range) |
|---|---|---|---|---|
| Development | Normal B cells | 12 | 2,293 | 140 (18–405) |
| Development | ALL | 30 | 4,815 | 112 (9–757) |
| External | Normal B cells | 14 | 1,096 | 41 (9–327) |
| External | ALL | 17 | 2,457 | 161 (14–278) |
| Total | | 73 | 10,661 | |

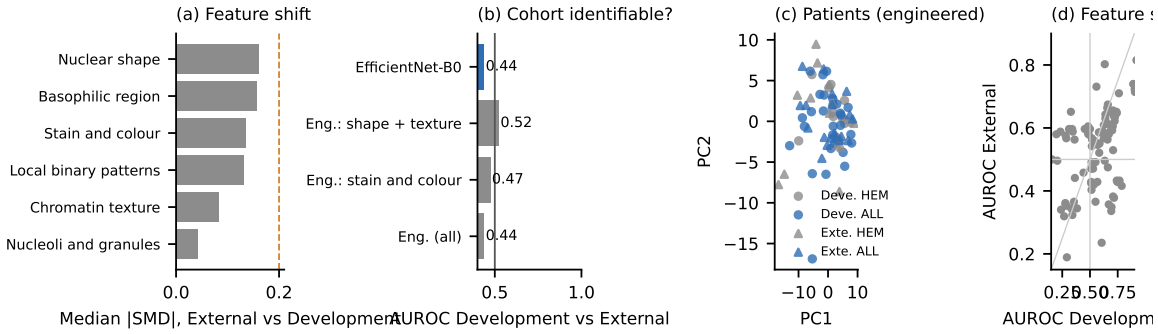



Fig. S2. Exploratory analysis of C-NMC (development: folds 0–1; external: fold 2), in the layout of the APL-AML analysis in the main text. (a) Shift of feature groups. (b) Set identifiability; values near 0.5 mean that external and development cells cannot be told apart. (c) Subjects on the first two principal components of their mean engineered features. (d) Subject-level AUROC of each feature in the two sets.

TABLE S4
LABEL-FREE SHIFT TEST. AUROC OF A CLASSIFIER THAT ASSIGNS CONTROL-CLASS CELLS (AML OR NORMAL B CELLS, UP TO 6,000) TO THE EXTERNAL OR DEVELOPMENT SET, PATIENT-GROUPED 5-FOLD CV. NULL: COHORT LABELS PERMUTED BETWEEN PATIENTS. THE TEST READS UNLABELLED EXTERNAL IMAGES AND IS THEREFORE TRANSDUCTIVE; IT USES NO EXTERNAL DIAGNOSES.

| Dataset | Representation | AUROC | Null mean (95th pct.) | $p$ | Perm. |
|---|---|---|---|---|---|
| APL-AML | Engineered: stain and colour | 0.753 | 0.498 (0.669) | $\leq$0.005 | 200 |
| | Engineered (all) | 0.704 | 0.486 (0.623) | 0.010 | 200 |
| | EfficientNet-B0 (frozen) | 0.583 | 0.479 (0.555) | 0.039 | 50 |
| | EfficientNet-B0 on Macenko images | 0.603 | 0.494 (0.590) | 0.059 | 50 |
| C-NMC | Engineered: stain and colour | 0.473 | 0.458 (0.682) | 0.448 | 200 |
| | Engineered (all) | 0.438 | 0.453 (0.712) | 0.542 | 200 |
| | EfficientNet-B0 (frozen) | 0.437 | 0.484 (0.647) | 0.706 | 50 |

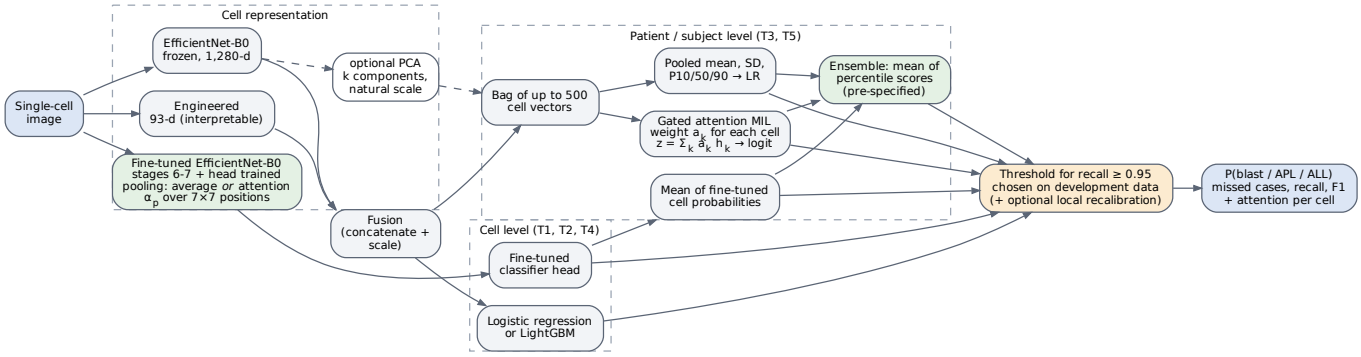



Fig. S3. Model overview. Each cell is described by engineered features, a frozen CNN embedding (optionally reduced by PCA), their fusion, or a partially fine-tuned CNN. Cell-level classifiers serve T1, T2 and T4. For patients and subjects (T3, T5), cells are combined by pooled statistics, by gated attention MIL or by averaging fine-tuned cell probabilities, and the three are combined in the pre-specified ensemble. The decision threshold is set on development data for a recall of at least 0.95.

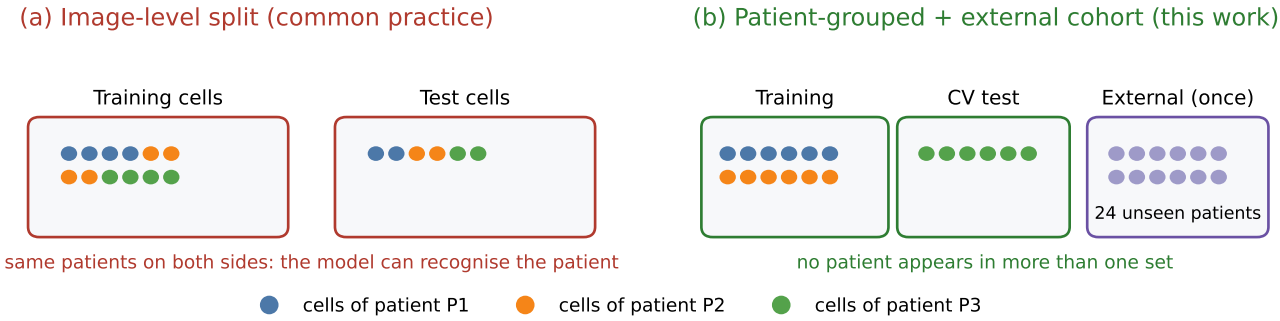



Fig. S4. Evaluation protocols. (a) Image-level splitting places cells of the same patient in training and test sets. (b) Patient-grouped splitting with an external cohort, used in this work; the external cohort is never used for fitting, threshold selection or calibration.

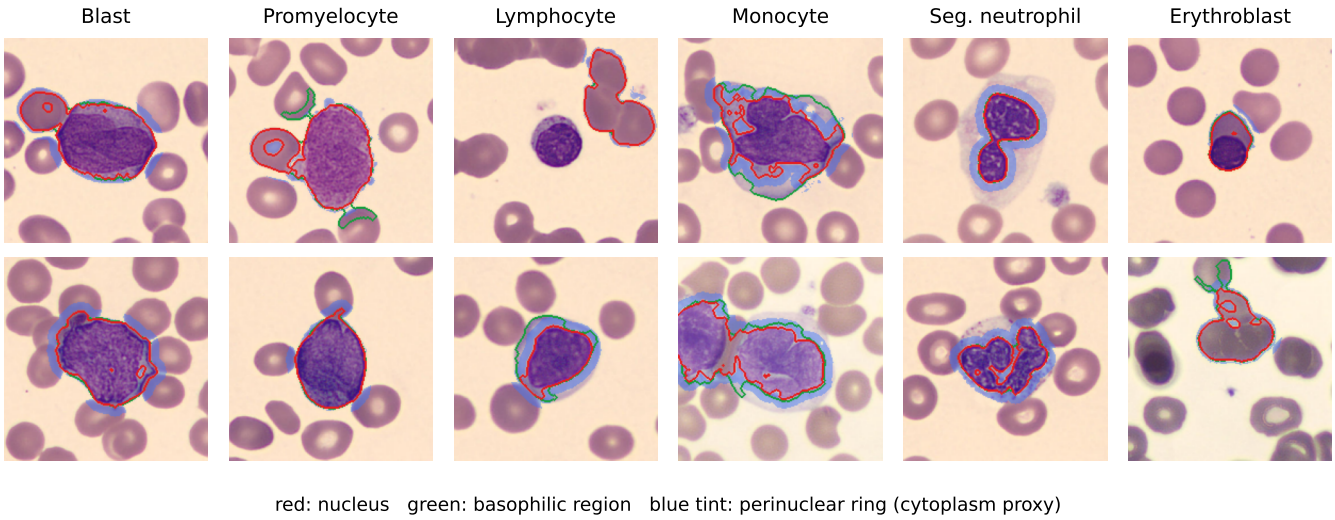



Fig. S5. Segmentation behind the engineered features, shown on cells at the 25th and 75th percentile of each class list. Red: nucleus; green: basophilic region; blue tint: perinuclear ring. The upper lymphocyte and lower erythroblast illustrate failures in which the mask falls on neighboring red cells.

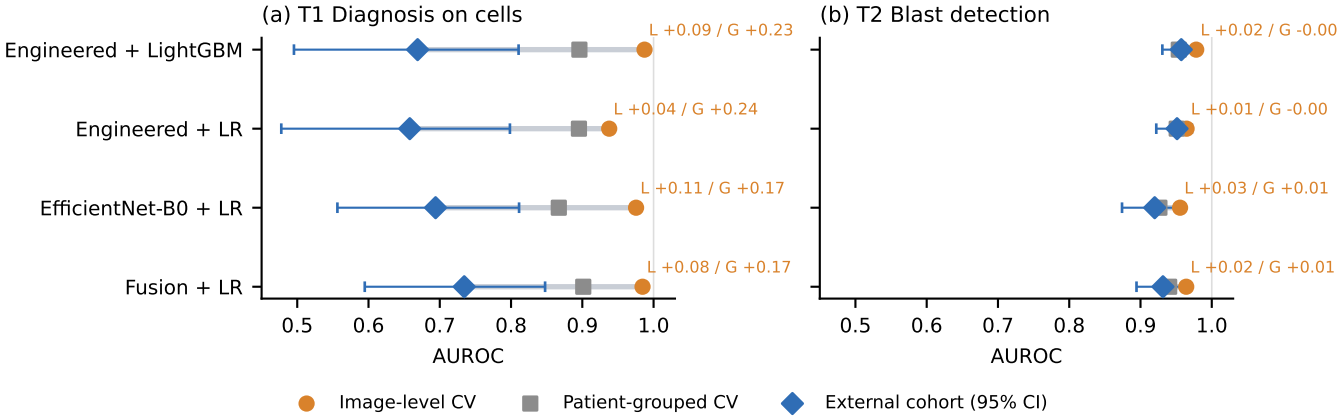



Fig. S6. AUROC under the three protocols at cell level (mean of three seeds). Orange: leakage (L, image-level minus patient-grouped CV) and external generalization gap (G, grouped CV minus external). (a) Cells labeled with the patient's diagnosis. (b) Cells labeled with their own type.

## S3. Feature engineering and full PCA sweep

TABLE S5

Feature engineering for patient-level APL detection. All steps are fitted inside the Discovery training folds; the Validation cohort was not used for fitting. PCA rows in this table use the whitened scaling of our first analysis (each component standardized), and the pooled LR of these experiments standardizes each feature, so its unreduced references differ from Table S8 and from the natural-scale analysis of the main text (engineered 0.729 vs. 0.757; EfficientNet-B0 0.743 vs. 0.722 and 0.708). Δ: external AUROC change relative to the unreduced representation (paired patient bootstrap). The configuration with the highest Discovery CV AUROC (pre-declared selection rule) is marked ⋆. Oracle column: specificity at 100% recall if the threshold were re-set on the Validation cohort (reference only, not deployable).

| Representation | Engineering step | CV AUROC | External AUROC [95% CI] | Δ [95% CI] | Recall (missed) | Spec. | F1 | Oracle spec. |
|---|---|---|---|---|---|---|---|---|
| Engineered, all 93 | – | 0.926 | 0.729 [0.511, 0.924] | – | 0.58 (5/12) | 0.67 | 0.61 | 0.33 |
| Engineered | Correlation filter ($\|r\| > 0.9$) | 0.922 | 0.736 [0.529, 0.930] | +0.007 [-0.070, +0.084] | 0.58 (5/12) | 0.67 | 0.61 | 0.42 |
| Engineered | Per-slide stain normalization | 0.872 | 0.576 [0.329, 0.818] | -0.153 [-0.326, +0.021] | 0.67 (4/12) | 0.50 | 0.62 | 0.08 |
| Engineered | L1 (LASSO) selection | 0.898 | 0.590 [0.350, 0.820] | -0.139 [-0.274, -0.023]* | 0.75 (3/12) | 0.42 | 0.64 | 0.17 |
| Engineered | Filter + stain normalization | 0.869 | 0.569 [0.322, 0.798] | -0.160 [-0.333, +0.009] | 0.67 (4/12) | 0.50 | 0.62 | 0.08 |
| EfficientNet-B0, 1,280-d | – | 0.957 | 0.743 [0.538, 0.926] | – | 0.50 (6/12) | 0.67 | 0.55 | 0.42 |
| EfficientNet-B0 | PCA, 16 components | 0.909 | 0.792 [0.580, 0.963] | +0.049 [-0.096, +0.213] | 0.92 (1/12) | 0.67 | 0.81 | 0.42 |
| EfficientNet-B0 | PCA, 32 components | 0.958 | 0.833 [0.644, 0.972] | +0.090 [-0.014, +0.210] | 0.50 (6/12) | 0.92 | 0.63 | 0.42 |
| EfficientNet-B0⋆ | PCA, 64 components | 0.966 | 0.819 [0.604, 1.000] | +0.076 [-0.171, +0.324] | 0.50 (6/12) | 0.92 | 0.63 | 0.00 |
| Fusion | Filter + PCA-32 | 0.960 | 0.764 [0.566, 0.938] | +0.021 [-0.105, +0.140] | 0.58 (5/12) | 0.83 | 0.67 | 0.42 |
| Digital differential | 8 predicted cell classes | 0.852 | 0.785 [0.570, 0.953] | +0.056 [-0.104, +0.229] | 1.00 (0/12) | 0.33 | 0.75 | 0.50 |
| Differential + engineered | Filtered | 0.928 | 0.764 [0.562, 0.943] | +0.035 [-0.015, +0.111] | 0.58 (5/12) | 0.67 | 0.61 | 0.50 |
| Engineered, attention MIL | Filter + stain normalization | 0.922 | 0.569 [0.326, 0.793] | -0.160 [-0.422, +0.091] | 0.42 (7/12) | 0.75 | 0.50 | 0.17 |
| EfficientNet-B0, attention MIL | PCA-32 | 0.924 | 0.799 [0.580, 0.972] | +0.069 [-0.133, +0.281] | 0.50 (6/12) | 0.75 | 0.57 | 0.50 |

Recall, specificity and F1 at the operating point fixed on Discovery (target sensitivity 0.95). * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction across all comparisons in the table.

TABLE S6

Patient/subject-level detection before and after PCA of the EfficientNet-B0 embeddings (pooled-statistics LR), on both datasets. PCA is fitted on training-fold cells only; the principal-component scores keep their relative variances (no whitening). Std.: external AUROC when each component is instead standardized to unit variance before the classifier (whitening), as in our first analysis. Δ: change in external AUROC relative to no PCA under the same scaling (paired bootstrap). Recall, specificity at the operating point fixed on development data. Components 16, 32 and 64 were pre-specified.

| | APL-AML (24 external patients, 12 APL) | | | | | | C-NMC (31 external subjects, 17 ALL) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Components | Var. | Std. | External [95% CI] | Δ | Recall (missed) | Spec. | Var. | Std. | External [95% CI] | Δ | Recall (missed) | Spec. |
| None (1,280) | 1.00 | 0.743 | 0.708 [0.493, 0.907] | – | 0.67 (4/12) | 0.67 | 1.00 | 0.908 | 0.908 [0.786, 0.992] | – | 1.00 (0/17) | 0.64 |
| 8 | 0.35 | 0.674 | 0.646 [0.412, 0.860] | -0.062 | 0.33 (8/12) | 0.75 | 0.47 | 0.849 | 0.803 [0.637, 0.934] | -0.105* | 0.88 (2/17) | 0.36 |
| 16 | 0.45 | 0.792 | 0.715 [0.496, 0.907] | +0.007 | 0.50 (6/12) | 0.75 | 0.58 | 0.798 | 0.819 [0.657, 0.949] | -0.088* | 0.94 (1/17) | 0.36 |
| 32 | 0.55 | 0.833 | 0.785 [0.580, 0.950] | +0.076 | 0.50 (6/12) | 0.75 | 0.67 | 0.790 | 0.840 [0.689, 0.958] | -0.067 | 0.94 (1/17) | 0.43 |
| 64 | 0.64 | 0.819 | 0.792 [0.594, 0.957] | +0.083 | 0.50 (6/12) | 0.75 | 0.76 | 0.739 | 0.853 [0.705, 0.965] | -0.055 | 0.94 (1/17) | 0.50 |
| 128 | 0.73 | 0.646 | 0.778 [0.586, 0.950] | +0.069 | 0.50 (6/12) | 0.75 | 0.84 | 0.676 | 0.866 [0.726, 0.971] | -0.042 | 1.00 (0/17) | 0.50 |
| 256 | 0.83 | 0.743 | 0.778 [0.586, 0.950] | +0.069 | 0.50 (6/12) | 0.75 | 0.91 | 0.655 | 0.870 [0.732, 0.974] | -0.038 | 1.00 (0/17) | 0.50 |

Var.: cumulative explained variance on development cells. * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction.

## S4. Training settings and learning curves

TABLE S7

Attention MIL trained for a fixed 40 epochs or with the number of epochs chosen by nested 3-fold CV inside each training fold (lowest mean validation loss). Development CV: range of the AUROC over three seeds. External AUROC, F1 at the development threshold and calibration error (ECE). Chosen epochs: over all folds and seeds.

| | | Dev. CV AUROC (seed range) | | External AUROC | | F1 | | ECE | | Chosen epochs |
|---|---|---|---|---|---|---|---|---|---|---|
| Dataset | Features | Fixed | Nested | Fixed | Nested | Fixed | Nested | Fixed | Nested | median (range) |
| APL-AML | Engineered | 0.824–0.830 | 0.770–0.852 | 0.812 | 0.819 | 0.67 | 0.67 | 0.26 | 0.26 | 18 (3–37) |
| | EfficientNet-B0 | 0.872–0.910 | 0.880–0.946 | 0.674 | 0.667 | 0.55 | 0.57 | 0.36 | 0.36 | 5 (3–7) |
| | Fusion | 0.859–0.930 | 0.869–0.912 | 0.681 | 0.688 | 0.55 | 0.55 | 0.33 | 0.35 | 5 (2–8) |
| C-NMC | Engineered | 0.864–0.911 | 0.861–0.881 | 0.912 | 0.895 | 0.81 | 0.81 | 0.25 | 0.27 | 33 (15–40) |
| | EfficientNet-B0 | 0.858–0.894 | 0.810–0.861 | 0.853 | 0.798 | 0.81 | 0.74 | 0.20 | 0.22 | 4 (1–36) |
| | Fusion | 0.854–0.911 | 0.824–0.897 | 0.853 | 0.811 | 0.85 | 0.76 | 0.21 | 0.17 | 4 (1–40) |

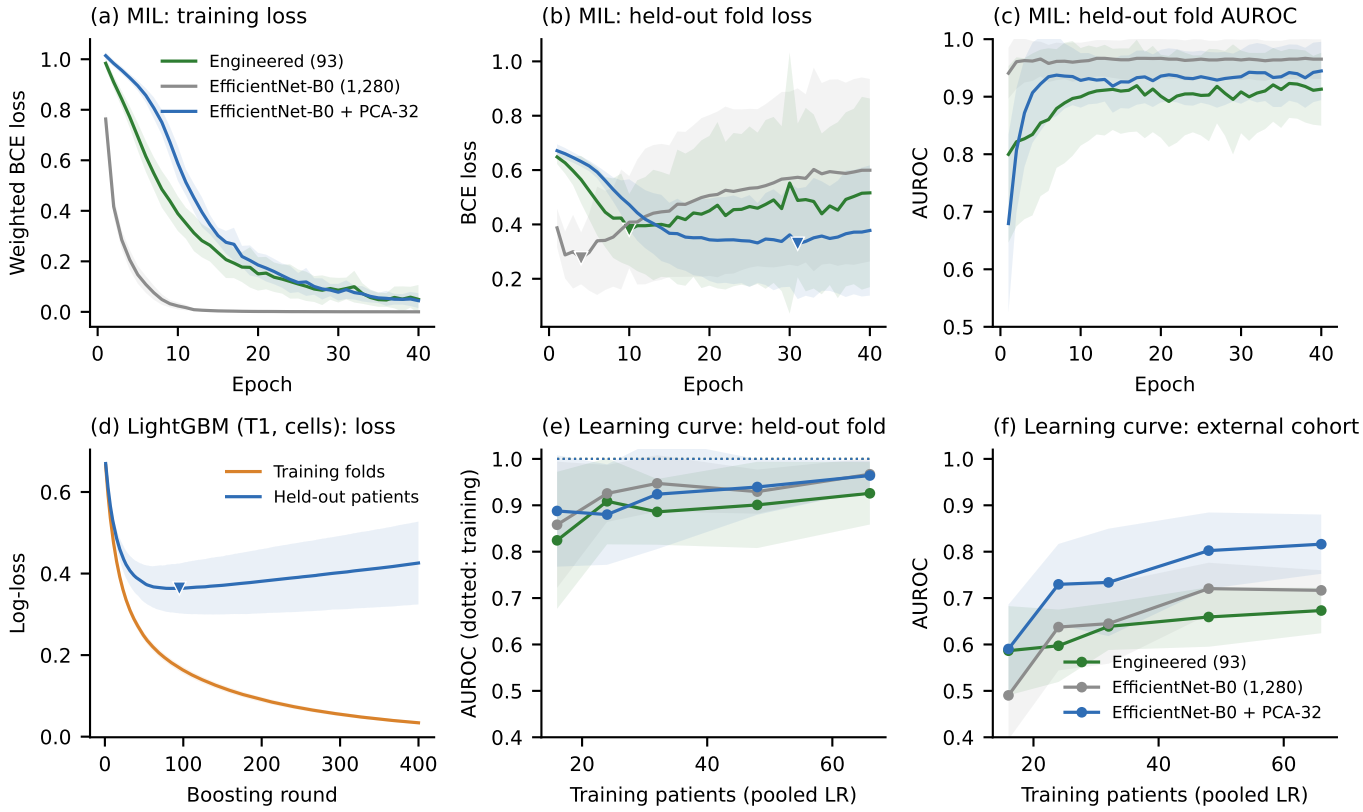



Fig. S7. Training dynamics on the Discovery cohort (mean ± SD over 5 folds × 3 seeds), computed in our first analysis with whitened PCA inputs (each component standardized); the main text uses natural-scale PCA. (a)–(c) Gated attention MIL trained with AdamW at a constant learning rate of $2 \times 10^{-4}$: training loss (class-weighted binary cross-entropy), loss and AUROC on the held-out fold per epoch; triangles mark the epoch with the lowest held-out loss. (d) Cell-level LightGBM for T1 (learning rate 0.05): log-loss on training folds and on held-out patients per boosting round. (e), (f) Learning curves of the pooled patient model versus the number of training patients, on the held-out fold (dotted: training AUROC) and on the external cohort. Values in (e) and (f) are for single fold models, not the fold ensemble used in the tables, and are therefore slightly lower.

## S5. Patient-level models without feature engineering

TABLE S8
Patient-level APL detection (T3). CV: Discovery, mean ± SD over three seeds. External: Validation cohort (12 APL, 12 AML) at the operating point fixed on Discovery (target sensitivity 0.95); the last three columns use a Youden threshold for reference.

| | | | External, sensitivity-first operating point | | | | | | External, Youden threshold | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Features | Aggregation | CV AUROC | AUROC [95% CI] | Recall (missed) | Spec. | F1 | NPV | ECE | Recall (missed) | Spec. | Acc. |
| Engineered | Pooled + LR | 0.867±0.059 | 0.757 [0.555, 0.938] | 1.00 (0/12) | 0.17 | 0.71 | 1.00 | 0.294 | 0.42 (7/12) | 0.92 | 0.67 |
| EfficientNet-B0 | Pooled + LR | 0.909±0.018 | 0.722 [0.493, 0.914] | 0.50 (6/12) | 0.67 | 0.55 | 0.57 | 0.338 | 0.50 (6/12) | 0.67 | 0.58 |
| Fusion | Pooled + LR | 0.910±0.018 | 0.708 [0.486, 0.909] | 0.50 (6/12) | 0.67 | 0.55 | 0.57 | 0.339 | 0.50 (6/12) | 0.67 | 0.58 |
| Engineered | Attention MIL | 0.827±0.003 | **0.812 [0.614, 0.963]** | 1.00 (0/12) | 0.00 | 0.67 | n/a | 0.264 | 0.50 (6/12) | 0.75 | 0.62 |
| EfficientNet-B0 | Attention MIL | 0.885±0.021 | 0.674 [0.444, 0.879] | 0.50 (6/12) | 0.67 | 0.55 | 0.57 | 0.362 | 0.50 (6/12) | 0.75 | 0.62 |
| Fusion | Attention MIL | 0.886±0.039 | 0.681 [0.445, 0.879] | 0.50 (6/12) | 0.67 | 0.55 | 0.57 | 0.325 | 0.50 (6/12) | 0.75 | 0.62 |

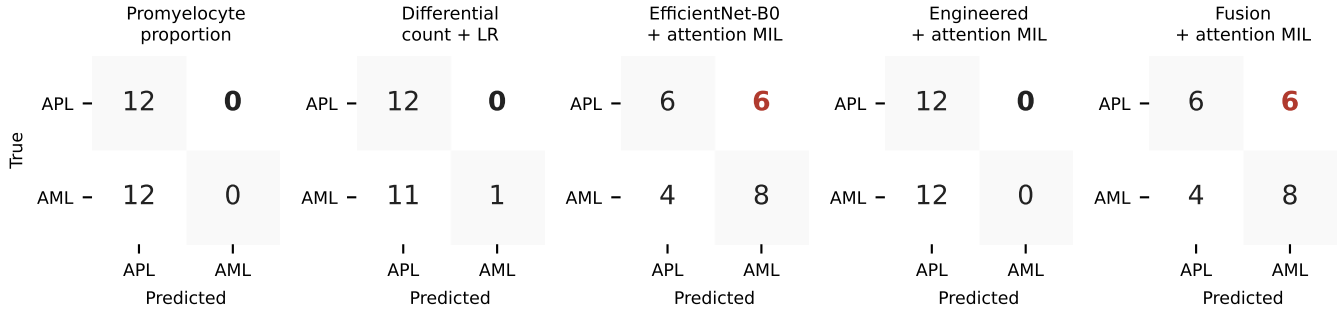



Fig. S8. Confusion matrices on the external cohort at the operating point fixed on Discovery for a sensitivity of at least 0.95. Missed APL patients are shown in red.

## S6. Pooled 106-patient analysis, local threshold adaptation and calibration

TABLE S9
SECONDARY ANALYSIS WITH MORE STATISTICAL POWER: REPEATED (3×) 5-FOLD PATIENT-LEVEL CV OVER ALL 106 APL-AML PATIENTS. THIS IS AN INTERNAL ANALYSIS, NOT AN EXTERNAL VALIDATION. Δ: AUROC DIFFERENCE VS. THE UNREDUCED EFFICIENTNET-B0 EMBEDDING (PAIRED PATIENT BOOTSTRAP ON REPEAT-AVERAGED SCORES). RECALL, SPECIFICITY AND F1 AT A THRESHOLD CHOSEN FOR SENSITIVITY $\geq 0.95$ INSIDE EACH TRAINING FOLD.

| Model | AUROC [95% CI] | Δ vs. no PCA | Recall (missed/34) | Spec. | F1 |
|---|---|---|---|---|---|
| Promyelocyte fraction (biomarker) | 0.677 [0.572, 0.784] | -0.275*† | 1.00 (0.0) | 0.00 | 0.49 |
| Engineered, pooled LR | 0.893 [0.823, 0.949] | -0.059* | 0.94 (2.0) | 0.42 | 0.59 |
| EfficientNet-B0, no PCA | 0.953 [0.898, 0.989] | – | 0.95 (1.7) | 0.53 | 0.64 |
| EfficientNet-B0 + PCA-16 | 0.878 [0.802, 0.941] | -0.074*† | 0.98 (0.7) | 0.35 | 0.58 |
| EfficientNet-B0 + PCA-32 | 0.947 [0.897, 0.985] | -0.005 | 0.96 (1.3) | 0.60 | 0.69 |
| EfficientNet-B0 + PCA-64 | 0.942 [0.892, 0.979] | -0.011 | 0.95 (1.7) | 0.63 | 0.70 |
| Engineered + PCA-32 fusion | 0.922 [0.853, 0.973] | -0.030 | 0.95 (1.7) | 0.58 | 0.67 |
| Engineered, attention MIL | 0.867 [0.794, 0.931] | -0.086* | 0.89 (3.7) | 0.52 | 0.62 |
| EfficientNet-B0, no PCA‡ | 0.929 [0.870, 0.974] | – | 0.96 (1.3) | 0.48 | 0.63 |
| EfficientNet-B0 + PCA-16‡ | 0.872 [0.791, 0.939] | -0.057* | 0.96 (1.3) | 0.31 | 0.56 |
| EfficientNet-B0 + PCA-32‡ | 0.929 [0.866, 0.976] | +0.000 | 0.96 (1.3) | 0.46 | 0.62 |
| EfficientNet-B0 + PCA-64‡ | 0.935 [0.876, 0.979] | +0.006 | 0.97 (1.0) | 0.49 | 0.63 |

* $p < 0.05$ unadjusted; † after Holm correction. ‡ PCA scores at natural scale; Δ vs. no PCA with the same scaling.

TABLE S10
LOCAL THRESHOLD ADAPTATION ON THE APL-AML VALIDATION COHORT (EXPLORATORY): $k$ CONFIRMED LOCAL PATIENTS ($k/2$ APL, $k/2$ AML) ARE DRAWN FROM THE VALIDATION COHORT AND THE REMAINING $24 - k$ PATIENTS ARE SCORED (2,000 RANDOM DRAWS). R0: DEVELOPMENT THRESHOLD. R2: DEVELOPMENT THRESHOLD LOWERED, IF NEEDED, TO JUST BELOW THE LOWEST SCORE OF THE LOCAL APL CASES. P(MISS): PROBABILITY THAT AT LEAST ONE APL PATIENT IN THE TEST SET IS MISSED.

| | R0, $k = 0$ | | | | R2, $k = 4$ | | | | R2, $k = 8$ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Model | Recall | Spec. | F1 | P(miss) | Recall | Spec. | F1 | P(miss) | Recall | Spec. | F1 | P(miss) |
| Macenko + frozen CNN, no PCA | 0.92 | 0.58 | 0.79 | 1.00 | 0.92 | 0.50 | 0.76 | 0.83 | 0.92 | 0.42 | 0.74 | 0.68 |
| Secondary: Macenko + frozen CNN + PCA-64 | 0.92 | 0.50 | 0.76 | 1.00 | 0.92 | 0.46 | 0.75 | 0.82 | 0.92 | 0.42 | 0.74 | 0.66 |
| Engineered, pooled LR | 1.00 | 0.17 | 0.71 | 0.00 | 1.00 | 0.17 | 0.71 | 0.00 | 1.00 | 0.17 | 0.71 | 0.00 |
| Engineered, attention MIL | 1.00 | 0.00 | 0.67 | 0.00 | 1.00 | 0.00 | 0.67 | 0.00 | 1.00 | 0.00 | 0.67 | 0.00 |
| Macenko + fine-tuned EfficientNet-B0 | 0.83 | 0.50 | 0.71 | 1.00 | 0.85 | 0.43 | 0.70 | 0.84 | 0.86 | 0.38 | 0.69 | 0.67 |
| **Primary: pre-specified ensemble** | 1.00 | 0.17 | 0.71 | 0.00 | 1.00 | 0.17 | 0.71 | 0.00 | 1.00 | 0.17 | 0.71 | 0.00 |

TABLE S11

CALIBRATION OF PATIENT-LEVEL PROBABILITIES ON THE EXTERNAL SETS. RAW: MODEL OUTPUT (NOT DEFINED FOR THE PERCENTILE-BASED ENSEMBLE). PLATT: LOGISTIC MAPPING FITTED ON DEVELOPMENT OUT-OF-FOLD SCORES. INTERCEPT: CALIBRATION-IN-THE-LARGE (0 = IDEAL; POSITIVE = RISK UNDERESTIMATED); SLOPE: 1 = IDEAL, $< 1$ = OVERCONFIDENT. PREVALENCE DIFFERS BETWEEN DEVELOPMENT AND EXTERNAL SETS (APL-AML 27% VS. 50%; C-NMC 71% VS. 55%), WHICH ALONE SHIFTS THE INTERCEPT.

| | | | Raw output | | | Platt (development) | | |
|---|---|---|---|---|---|---|---|---|
| Dataset | Model | AUROC | Brier | Intercept | Slope | Brier | Intercept | Slope |
| APL-AML | Engineered, pooled LR | 0.757 | 0.249 | 1.23 | 0.64 | 0.263 | 1.46 | 0.70 |
| | Engineered, attention MIL | 0.812 | 0.230 | 1.03 | 1.27 | 0.261 | 1.30 | 1.67 |
| | Macenko + fine-tuned CNN | 0.771 | 0.253 | 1.27 | 1.05 | 0.252 | 1.65 | 0.61 |
| | Secondary (Macenko + frozen CNN + PCA-64) | 0.639 | 0.295 | 1.07 | 0.34 | 0.316 | 1.42 | 0.31 |
| | Primary ensemble | 0.806 | – | – | – | 0.231 | 1.32 | 0.84 |
| C-NMC | Engineered, pooled LR | 0.941 | 0.136 | -0.74 | 2.34 | 0.147 | -1.47 | 1.53 |
| | Engineered, attention MIL | 0.912 | 0.213 | -1.58 | 2.63 | 0.200 | -1.68 | 1.75 |
| | Fine-tuned CNN | 0.962 | 0.160 | -0.77 | 5.40 | 0.159 | -1.09 | 3.98 |
| | Secondary (frozen CNN + PCA-64) | 0.853 | 0.178 | -1.28 | 0.79 | 0.184 | -1.29 | 0.99 |
| | Primary ensemble | 0.960 | – | – | – | 0.136 | -1.16 | 2.00 |

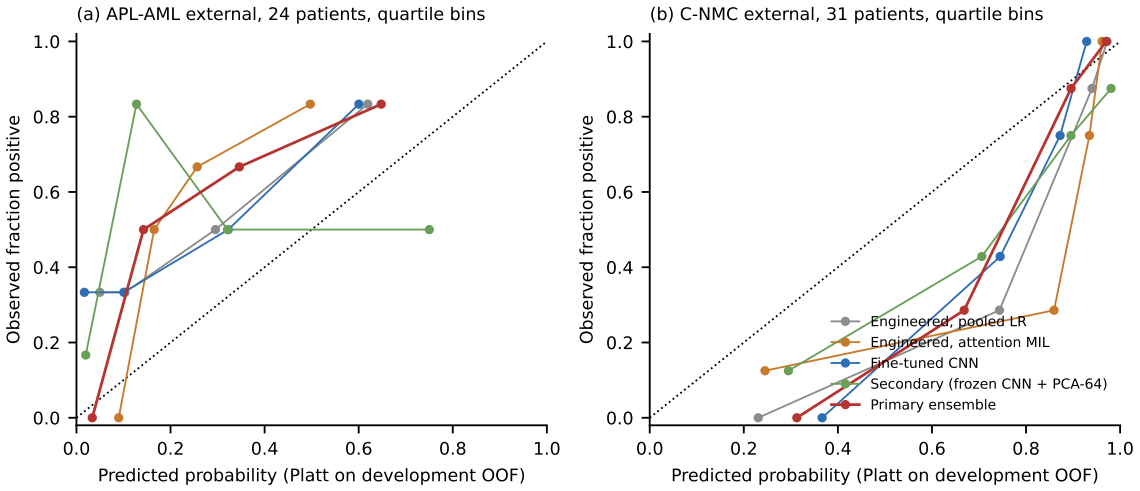



Fig. S9. Reliability diagrams of patient-level probabilities on the external sets after Platt scaling fitted on development out-of-fold scores (quartile bins; 24 and 31 patients, so each point rests on 6–8 patients). Dotted: perfect calibration.

TABLE S12

DOES THE NUMBER OF CELLS PER SUBJECT EXPLAIN C-NMC RESULTS? COUNT AUROC: AUROC OF THE CELL COUNT ALONE FOR ALL VS. NORMAL. FOR EACH MODEL: AUROC OF THE SUBJECT SCORE, ITS SPEARMAN CORRELATION WITH LOG(CELL COUNT), THE AUROC OF THE SCORE AFTER REGRESSING OUT LOG(COUNT), AND THE LIKELIHOOD-RATIO $p$-VALUE FOR THE SCORE ADDED TO LOG(COUNT).

| | Development (42) | External (31) |
|---|---|---|
| Quantity | Value | Value |
| Median cells per subject, ALL / normal | 112 / 140 | 161 / 41 |
| Count AUROC (Mann–Whitney $p$) | 0.453 (0.65) | 0.735 (0.028) |
| Model: AUROC / $\rho$(score, log count) / adjusted AUROC / $p$ | | |
| Engineered, pooled LR | 0.917 / -0.12 / 0.917 / <0.001 | 0.941 / +0.08 / 0.920 / <0.001 |
| Engineered, attention MIL | 0.856 / -0.21 / 0.944 / <0.001 | 0.912 / -0.03 / 0.950 / <0.001 |
| Fine-tuned EfficientNet-B0 | 0.864 / -0.29 / 0.892 / <0.001 | 0.962 / +0.15 / 0.933 / <0.001 |
| Primary ensemble | 0.886 / -0.20 / 0.911 / <0.001 | 0.960 / +0.06 / 0.950 / <0.001 |

TABLE S13

LOCAL THRESHOLD ADAPTATION WITH A SEPARATE CALIBRATION SAMPLE (INDEPENDENT COHORT; 2,000 RANDOM SPLITS). THRESHOLDS ARE SET ON THE CALIBRATION SAMPLE ONLY AND EVALUATED ON ALL REMAINING PATIENTS. POS: JUST BELOW THE LOWEST CALIBRATION APL SCORE; BOTH: THRESHOLD MAXIMIZING CALIBRATION SPECIFICITY SUBJECT TO CALIBRATION RECALL $\geq$0.9. P05: 5TH PERCENTILE OF EVALUATION RECALL OVER SPLITS; P(MISS): PROBABILITY OF MISSING AT LEAST ONE APL PATIENT IN THE EVALUATION SAMPLE.

| Model | Calibration (APL/AML) | Rule | Recall | P05 | Expected | Spec. | Referred | P(miss) |
|---|---|---|---|---|---|---|---|---|
| Primary pipeline | 5/20 | POS | 0.84 | 0.53 | 0.83 | 0.59 | 0.49 | 0.79 |
| Primary pipeline | 5/20 | BOTH | 0.84 | 0.53 | 0.83 | 0.59 | 0.49 | 0.79 |
| Primary pipeline | 8/30 | POS | 0.89 | 0.62 | 0.89 | 0.48 | 0.58 | 0.67 |
| Primary pipeline | 8/30 | BOTH | 0.89 | 0.62 | 0.89 | 0.48 | 0.58 | 0.67 |
| Primary pipeline | 12/50 | POS | 0.93 | 0.75 | 0.92 | 0.35 | 0.70 | 0.48 |
| Primary pipeline | 12/50 | BOTH | 0.86 | 0.58 | 0.92 | 0.69 | 0.41 | 0.76 |
| DinoBloom + engineered | 5/20 | POS | 0.83 | 0.47 | 0.83 | 0.59 | 0.49 | 0.82 |
| DinoBloom + engineered | 5/20 | BOTH | 0.83 | 0.47 | 0.83 | 0.59 | 0.49 | 0.82 |
| DinoBloom + engineered | 8/30 | POS | 0.89 | 0.62 | 0.89 | 0.45 | 0.61 | 0.67 |
| DinoBloom + engineered | 8/30 | BOTH | 0.89 | 0.62 | 0.89 | 0.45 | 0.61 | 0.67 |
| DinoBloom + engineered | 12/50 | POS | 0.93 | 0.75 | 0.92 | 0.34 | 0.70 | 0.49 |
| DinoBloom + engineered | 12/50 | BOTH | 0.85 | 0.58 | 0.92 | 0.59 | 0.49 | 0.77 |
| Engineered, pooled LR | 5/20 | POS | 0.83 | 0.53 | 0.83 | 0.55 | 0.52 | 0.79 |
| Engineered, pooled LR | 5/20 | BOTH | 0.83 | 0.53 | 0.83 | 0.55 | 0.52 | 0.79 |
| Engineered, pooled LR | 8/30 | POS | 0.89 | 0.62 | 0.89 | 0.43 | 0.62 | 0.67 |
| Engineered, pooled LR | 8/30 | BOTH | 0.89 | 0.62 | 0.89 | 0.43 | 0.62 | 0.67 |
| Engineered, pooled LR | 12/50 | POS | 0.92 | 0.75 | 0.92 | 0.31 | 0.73 | 0.51 |
| Engineered, pooled LR | 12/50 | BOTH | 0.84 | 0.58 | 0.92 | 0.62 | 0.46 | 0.76 |
| Fine-tuned CNN | 5/20 | POS | 0.83 | 0.53 | 0.83 | 0.44 | 0.61 | 0.78 |
| Fine-tuned CNN | 5/20 | BOTH | 0.83 | 0.53 | 0.83 | 0.44 | 0.61 | 0.78 |
| Fine-tuned CNN | 8/30 | POS | 0.89 | 0.62 | 0.89 | 0.36 | 0.69 | 0.68 |
| Fine-tuned CNN | 8/30 | BOTH | 0.89 | 0.62 | 0.89 | 0.36 | 0.69 | 0.68 |
| Fine-tuned CNN | 12/50 | POS | 0.93 | 0.75 | 0.92 | 0.27 | 0.76 | 0.50 |
| Fine-tuned CNN | 12/50 | BOTH | 0.85 | 0.58 | 0.92 | 0.45 | 0.60 | 0.75 |

TABLE S14

ANALYSIS REGISTRY: WHICH RESULTS ARE CONFIRMATORY AND WHICH EXPLORATORY, AND WHAT EACH TEST SET HAD BEEN EXPOSED TO BEFORE IT WAS SCORED.

| Analysis | Status | Exposure of the test set before scoring |
|---|---|---|
| Leakage vs. external generalization gap (T1, T2, T4) | Pre-planned, descriptive | External sets used only for scoring; same models and folds for all protocols |
| APL-AML later (Validation) cohort, all models | Exploratory | Evaluated repeatedly in the first analysis and the revisions; preprocessing subset included its patients (reused test set) |
| C-NMC fold 2, all models | Secondary | Evaluated for several configurations; never used for selection |
| Pooled 106-patient CV | Secondary, internal | Not an external test |
| Primary and secondary pipelines | Pre-specified (commit `f47f3ec`) | Specified after first-analysis results on the Validation cohort were known |
| Independent MLL cohort: primary, secondary, DinoBloom + engineered | **Confirmatory** (commit `acf173b`) | Untouched: plan, models and thresholds fixed before any MLL image was scored |
| Workload, split-sample recalibration, reciprocal transfer on MLL | Post hoc (requested in review) | Use MLL scores after the confirmatory analysis; no model changed |
| Scan-time confounding, shift tests, calibration, bag size | Exploratory | Label-free or development-only, except calibration on external scores |
| C-NMC improvement pilot (crop, LightGBM, DinoBloom) | Exploratory, development CV | External fold 2 reported after selection on development CV |

## S7. Ablation and explanation of engineered features

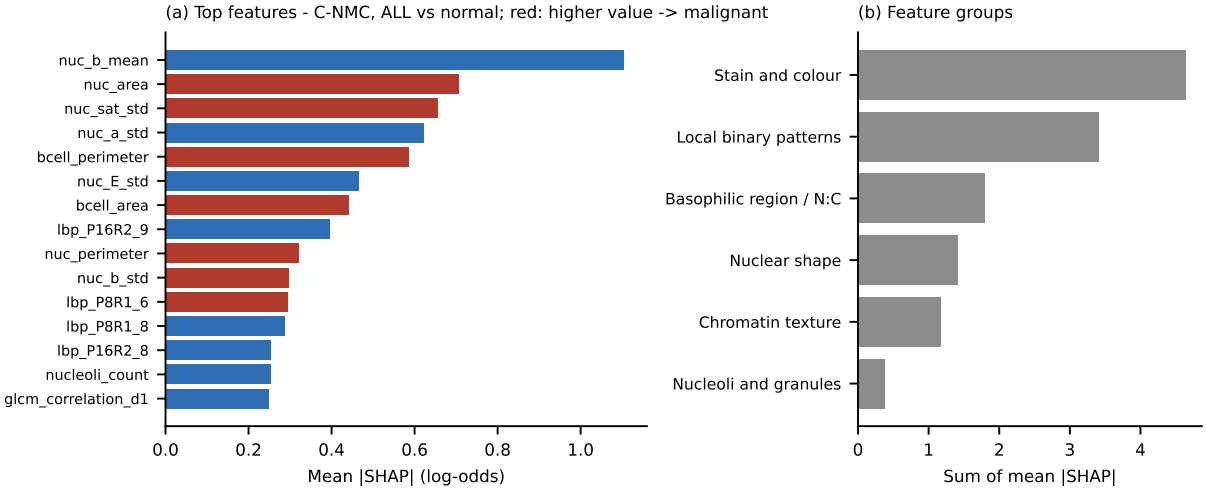



Fig. S10. SHAP explanation of the engineered-feature LightGBM cell classifier on C-NMC (trained on development cells, explained on the 3,553 external cells). (a) Fifteen most influential features (mean absolute SHAP value, log-odds); red: higher values push towards ALL, blue: towards normal. (b) Total contribution of each feature group.

TABLE S15
FEATURE-GROUP ABLATION ON THE EXTERNAL COHORT (ENGINEERED FEATURES, LR). Δ: CHANGE IN AUROC RELATIVE TO ALL 93 FEATURES, PAIRED PATIENT BOOTSTRAP.

| | T2 Blast detection | | T1 Diagnosis on cells | |
|---|---|---|---|---|
| Setting ($n$) | AUROC | Δ [95% CI] | AUROC | Δ [95% CI] |
| All features (93) | 0.950 | – | 0.671 | – |
| *Without one group* | | | | |
| Nuclear shape (84) | 0.949 | -0.001 [-0.003, +0.001] | 0.681 | +0.010 [-0.004, +0.024] |
| Basophilic region (83) | 0.945 | -0.005 [-0.009, -0.003]* | 0.665 | -0.005 [-0.025, +0.010] |
| Stain and colour (67) | 0.916 | -0.034 [-0.060, -0.015]* | 0.548 | -0.123 [-0.272, +0.046] |
| Chromatin texture (77) | 0.939 | -0.012 [-0.023, -0.004]* | 0.675 | +0.004 [-0.007, +0.016] |
| Local binary patterns (65) | 0.947 | -0.004 [-0.010, +0.002] | 0.670 | -0.001 [-0.019, +0.020] |
| Nucleoli and granules (89) | 0.951 | +0.001 [-0.002, +0.004] | 0.671 | +0.000 [-0.006, +0.006] |
| *One group only* | | | | |
| Nuclear shape (9) | 0.810 | -0.140 [-0.212, -0.082]* | 0.549 | -0.121 [-0.283, +0.062] |
| Basophilic region (10) | 0.823 | -0.127 [-0.198, -0.069]* | 0.579 | -0.092 [-0.242, +0.073] |
| Stain and colour (26) | 0.912 | -0.038 [-0.078, -0.013]* | 0.682 | +0.011 [-0.032, +0.041] |
| Chromatin texture (16) | 0.802 | -0.148 [-0.221, -0.086]* | 0.526 | -0.145 [-0.331, +0.067] |
| Local binary patterns (28) | 0.793 | -0.157 [-0.225, -0.098]* | 0.581 | -0.090 [-0.241, +0.081] |
| Nucleoli and granules (4) | 0.860 | -0.090 [-0.142, -0.054]* | 0.500 | -0.171 [-0.365, +0.057] |

* $p < 0.05$.

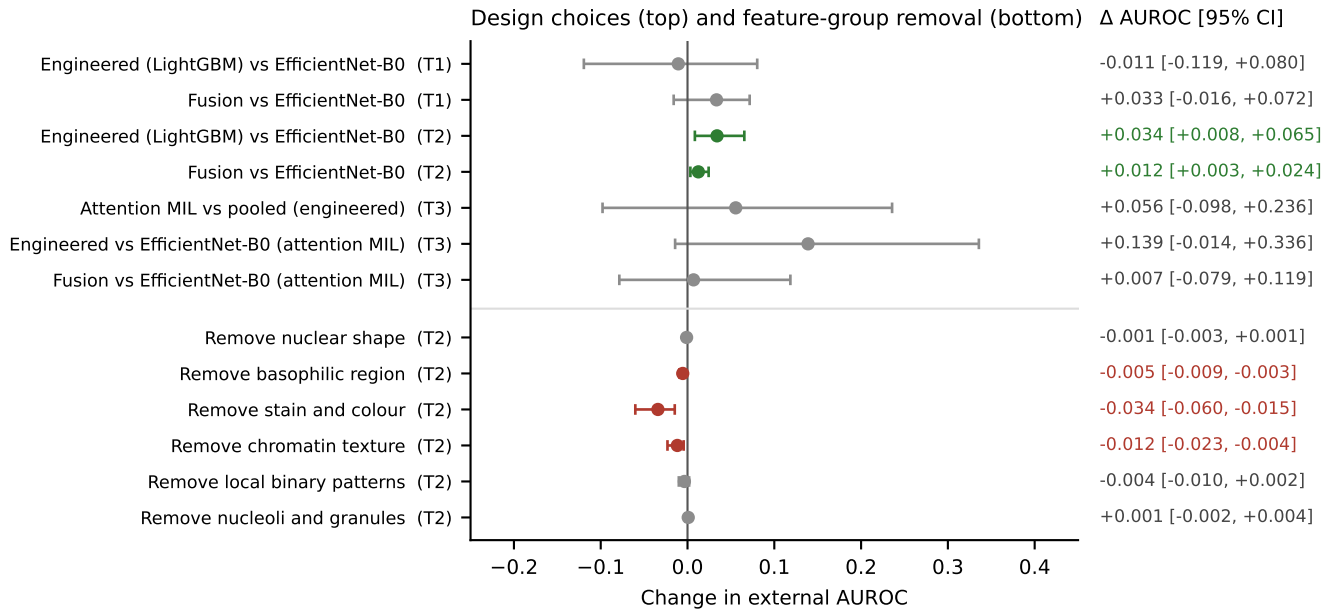



Fig. S11. Effect of design choices (top) and of removing each engineered feature group for blast detection (bottom) on external AUROC. Bars are 95% confidence intervals from a paired patient-level bootstrap.

## S8. TRIPOD+AI CHECKLIST

TABLE S16
TRIPOD+AI CHECKLIST [2] (ABBREVIATED ITEM WORDING). M: MAIN TEXT; S: THIS SUPPLEMENT; N/A: NOT APPLICABLE.

| Item | Topic | Where reported |
|---|---|---|
| 1 | Title: development/evaluation, population, outcome | M, title |
| 2 | Abstract | M, abstract |
| 3 | Background, rationale, existing models | M, Introduction; Related Work; Table I |
| 4 | Objectives | M, contributions C1–C5 |
| 5 | Data sources, study dates | M, Datasets (APL-AML 2010–2020, split January 2019; C-NMC 2019 folds; MLL cohort 2009–2020, prespecified evaluation) |
| 6 | Setting, eligibility, treatments | M, Datasets; all patients of both public datasets were used |
| 7 | Data preparation, quality checks | M, Preprocessing; 10 unreadable files kept as missing; Fig. SS5 |
| 8 | Outcome definition and blinding | M, Learning Tasks; diagnoses from the dataset providers; no new labels were created |
| 9 | Predictors (features) and their measurement | M, Feature Representations; Table S1 |
| 10 | Sample size | M, Datasets; small external sets discussed with [3], [4] |
| 11 | Missing data | M, Preprocessing (imputation inside training folds) |
| 12 | Analytical methods: model type, building, tuning, internal validation | M, Models; Pipelines; Evaluation Protocol; Table S1 |
| 13 | Class imbalance | M, Exploratory Analysis (class weights, no resampling, positive-class thresholds) |
| 14 | Fairness | n/a: no demographic variables are distributed with either dataset |
| 15 | Model output and thresholds | M, Pipelines; threshold for recall $\geq$0.95 on development data |
| 16 | Training vs. evaluation data differences | M, Exploratory Analysis; shift test (Table S4); scan-time confounding |
| 17 | Ethics | Public de-identified data; no new data collected |
| 18 | Funding, conflicts | To be completed by the authors |
| 19 | Protocol, registration | Analysis plans committed to the code repository before scoring (commits f47f3ec, acf173b); no formal registry |
| 20 | Data and code availability | M, Data and Code Availability |
| 21 | Patient and public involvement | None |
| 22 | Participants: flow, characteristics | M, Table II; Table S3; Fig. SS1 |
| 23 | Model performance with uncertainty (discrimination, calibration, clinical utility) | M, Results; patient bootstrap CIs; calibration (Table S11); decision curves |
| 24 | Model updating / recalibration | M, Local Threshold Adaptation |
| 25 | Interpretation, limitations | M, Discussion |
| 26 | Usability, intended use | M, Pipelines (prioritization for urgent review, not rule-out) |
| 27 | Next steps | M, Discussion and Conclusion |

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[4] G. S. Collins, E. O. Ogundimu, and D. G. Altman, "Sample size considerations for the external validation of a multivariable prognostic model: a resampling study," *Statistics in Medicine*, vol. 35, no. 2, pp. 214–226, 2016.